# Forecasting Next-Week Volatility of S&P 500 Stocks

## Milestone 2: data wrangling and project redefinition

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CS1090A / AC209A, Fall 2026  Group 38, project 12  TF: Daniel More Torres

# The question and why it matters



- Returns are close to unpredictable; **volatility clusters**, so it can be forecast.
- Volatility forecasts size positions, set risk limits and price options.
- HAR (Corsi, 2009), a regression on last day, week and month of volatility, is hard to beat.
- **Our question**: with free daily data, does one pooled model with cross-sectional features beat HAR fit stock by stock, out of sample, and in which market regimes?

# The data



| Source | Rows | One row is | Key → values | Access |
|---|---|---|---|---|
| S&P 500 [1] | 1.35M | a member on a date | $(d, i)$ | GitHub |
| FRED [2] | 5.7K | a market day | $d \to$ VIX, VIX3M, credit, slope, T-bill | Website |
| Yahoo Finance [3] | 3.21M | a stock's day | $(d, i) \to$ OHLCV | `yfinance` |
| SEC EDGAR [4] | 53K | an earnings filing | $(i, \tau) \to$ SIC code | REST API |
| Alpaca [5] | 1.72M | a stock's day$^{\dagger}$ | $(d, i) \to$ OHLCV, RV | REST API |
| CRSP[1] [6] | 3.75M | a stock's day$^{\dagger}$ | $(d, i) \to$ OHLCV | WRDS SQL |

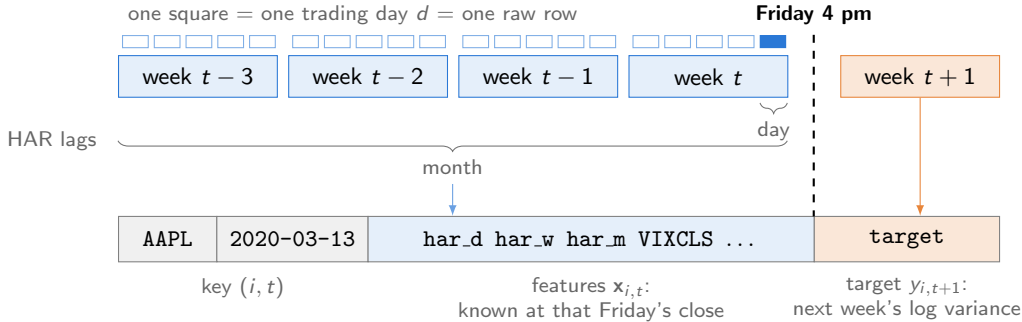
OHLCV: open, high, low, close, volume. RV: realized variance. SIC: industry code. $^{\dagger}$Incl. delisted stocks.

- **Notation**: $d$ a trading day; $i$ a stock ticker, e.g. AAPL; $\tau$ a filing time.
- **Range**: prices 2005–2026; Alpaca from 2016, CRSP to 2025; members since 1996.
- **Cached**: every fetch is cached, so the notebook needs no keys.

[1] CRSP is the reference price database in finance research, built at the University of Chicago; we query it through WRDS, Wharton's (UPenn) research data service, under Harvard's license.

# What one row is





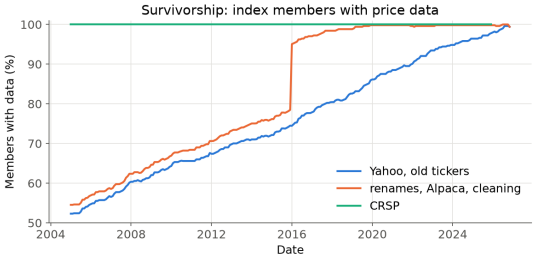

- **Raw table**: one row per $(d, i)$; 3,423,011 rows = Yahoo's 3.21M + Alpaca's 215K delisted.
- **Modeling table**: one row per $(i, t)$, $t$ a week ending Friday; 473,684 rows $\times$ 183 columns.

Learn $\mathbf{x}_{i,t} \longmapsto y_{i,t+1}$

the stock-by-period panel of Gu et al. [7]

# Who is missing: survivorship



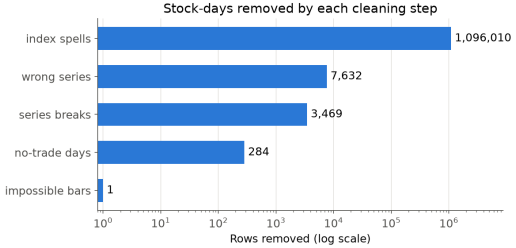



**Why check**: each week we forecast every index member, including firms that later leave; no one knows in advance which will.

- Yahoo keeps only firms still listed today: coverage 55% of 2005 members, 99.8% in 2026.
- Alpaca keeps delisted firms, but only from 2016: the jump.
- Before 2016, firms that left (acquired or bankrupt) are missing: not a random sample.
- CRSP check, 2005–2015: 191,083 vs 286,597 stock-weeks; the firms we keep are slightly calmer (mean log variance 0.04 lower).
- From 2019 under 1% missing: the test period is barely affected.

# Cleaning the raw prices: identity, then values






**Identity**: is each row the index member on that day?

1. **Wrong time**: tickers are reused (today's FB is an ETF), so keep rows near each index spell.
2. **Wrong company**: 9 series are another security (Yahoo's PARA near $100,000).
3. **Company switch**: 4 series change firm on a date (JCI before 2016 is Tyco).

**Values**: 1 impossible bar, 284 zero-volume days removed.

**Unusual values kept**: 80 days with high = low, so zero Garman–Klass variance (floored at $10^{-6}$); the largest daily returns are real events (Fannie Mae −90% in 2008, SVB −60% in 2023).

3,423,011 → 2,315,615 stock-days. No rule reads later rows.

# The modeling table: missing values and feature scale



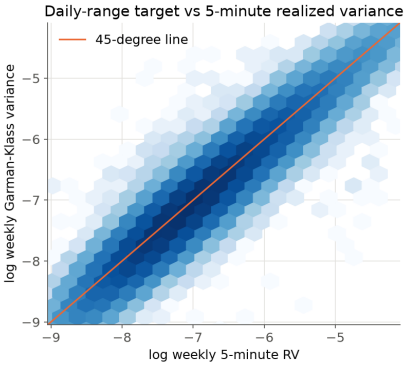
**Missing values**: no cell is missing in the clean prices; gaps appear once features are built.

| Column | Missing | Why |
|---|---|---|
| 3-month VIX (`VXVCLS`) | 9.2% | series starts December 2007 |
| earnings timing | 7–8% | delisted firms are absent from SEC's current ticker list |
| 60-day Alpha158 windows | 1.0% | first weeks after a firm enters the index |

- **No row is dropped.** Boosted trees take NaN; ridge and lasso impute the training-window median plus a missing-value indicator. Milestone 3 compares imputation choices.
- **Scale**: feature standard deviations run from 0.004 to 14, so ridge and lasso standardize every feature inside each training window; trees need no scaling.

# Volatility from daily prices lines up with 5-minute data

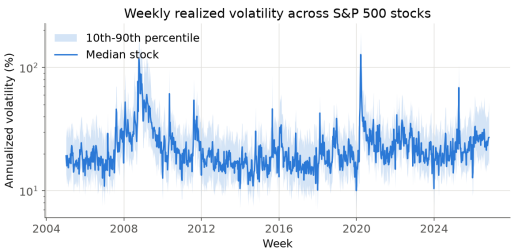





- **We predict** how much each stock will move next week (its variance).
- **We measure it** from each day's open, high, low and close (Garman–Klass, 1980): $\log \sum_{d \in \text{next week}} \left[ \frac{1}{2} \log^2 \frac{H_d}{L_d} - (2 \log 2 - 1) \log^2 \frac{C_d}{O_d} \right]$
- **We checked it** against 5-minute data for 697 companies: correlation 0.95, and 0.94 within a single company.
- **We leave out** the jump from one day's close to the next day's open (the overnight move), a median 27% of a stock's daily variance in 2005–2018.

# Volatility spikes in a few rare weeks (stress weeks)






- **Stress week:** VIX, the market's fear gauge, is 30 or more at the week's close, so we know it before we forecast.
- **How often:** 89 of 1,139 weeks, mostly 2008–09, 2011, 2020 and 2022.
- **In the test years:** 29 stress weeks, 16 of them in 2020. The tuning years have 1.

# How we keep future data out of the forecast (leakage)



| Risk | Control | Enforced by |
|---|---|---|
| a feature sees next week's prices | each feature uses only days up to Friday's 4 pm close | `features.py` test: adding later weeks changes no earlier value |
| earnings news after the 4 pm close | an earnings filing (8-K) after 4 pm counts for the next week | `features.py` test: a 4:30 pm Friday filing counts next week |
| picking stocks from today's S&P 500 list | each date uses that day's index members | `clean.py` test: a stock counts only between entry and exit |
| VIX closes at 4:15 pm, after stocks | Friday's 4:15 pm close is used only for next week | `features.py` test: each week gets that Friday's VIX |
| training weeks run into the test weeks | a one-week gap, as each target looks a week ahead | not yet: the split code comes in Milestone 3 |

Still in progress: checking when each data source first published its values.

# Revised question



Train 2005–2015, tune 2016–2018, test 2019–2026 with yearly expanding refits.

1. **Accuracy**: do pooled ridge/lasso and gradient-boosted trees beat per-stock HAR on mean QLIKE and MSE? Diebold–Mariano at 5% on the weekly mean loss difference (Newey–West); all models scored on the same rows. A second baseline: HAR pooled across stocks with a constant per stock, to separate the gain from pooling from the gain from features. QLIKE on the exponential of a log forecast gets a training-only smearing correction (Duan, 1983).

2. **Attribution and regimes**: which feature blocks carry any gain (refit without each block), and does it keep its sign in calm and in stress weeks? Holm correction across the family of tests.

**Changed**: economic-value backtest dropped; earnings from 8-K timestamps; fundamentals deferred (XBRL starts 2009); Alpha158 added; stress defined by VIX, not calendar years.

# Staff notes and the AC209A requirement



| Staff note | What we do |
|---|---|
| omit economic value, Sharpe, drawdown, factor regression | dropped entirely |
| prioritize regularized regression and boosted trees | the two pooled models of Q1 |
| clustering and neural networks optional | only if time remains after Q1 and Q2 |
| yfinance can fail; delisted stocks missing | cached fetches; renames, wrong series and Alpaca fill; coverage reported |
| XBRL only from mid-2009 | fundamentals deferred |
| AC209A: one method not covered in class | HAR, GARCH(1,1), QLIKE loss (Patton, 2011), Diebold–Mariano test |

# Next steps and questions for you



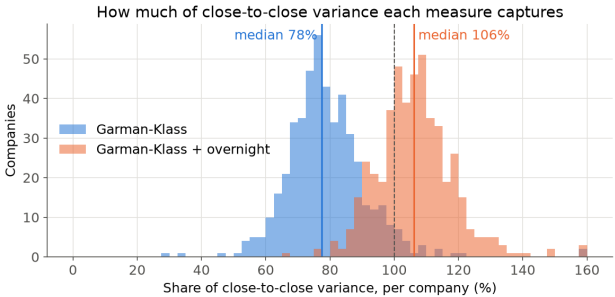
**Milestone 3** (proposed owners)

- HAR (per stock, and pooled with a constant per stock) and GARCH baselines: Simon
- ridge and lasso: Shuo
- gradient-boosted trees: Ashley
- QLIKE, MSE, Diebold–Mariano: Terry
- data upkeep and CRSP: Juntang

**Questions**

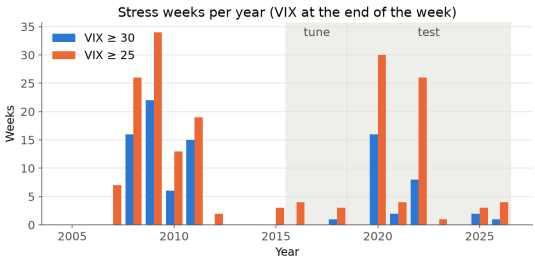
1. May Milestone 3 train on CRSP (WRDS) to 2025 and use free data for 2026?
2. Do HAR, GARCH, QLIKE and Diebold–Mariano meet the AC209A requirement?
3. Yearly expanding refits on 2019–2026, or one fit?
4. Target: trading-hours variance as now, or add the overnight term back?

# Backup: what the daily range misses




- 477 companies, 2005–2018 only (train and tune years), days inside the index spell. Each share sums a company's days, so 2008–09 weighs most.
- Overnight is a median 27% of close-to-close variance (19% to 37%, 10th to 90th percentile): 25% in 2005–09 and 30% in 2010–18 for the 317 companies in both.
- Earnings released after the close show up in this move.
- `rv_overnight` is already a column in the weekly table.

# Backup: stress weeks by split




| Split | Weeks | VIX $\geq 30$ | episodes | VIX $\geq 25$ |
|---|---|---|---|---|
| train | 573 | 59 | 8 | 104 |
| tune | 157 | 1 | 1 | 7 |
| test | 409 | 29 | 13 | 68 |

- Tuning (2016–18) has 1 stress week at VIX $\geq 30$.
- 16 of the test's 29 stress weeks are in 2020.

# References


[1] Farrell Aultman. sp500: Current and historical lists of S&P 500 components since 1996. https://github.com/fja05680/sp500, 2026. GitHub repository, commit a2430f2 of 2026-09-07.

[2] Federal Reserve Bank of St. Louis. FRED, Federal Reserve Economic Data. https://fred.stlouisfed.org, 2026.

[3] Ran Aroussi. yfinance: Download market data from Yahoo! Finance's API. https://github.com/ranaroussi/yfinance, 2026. Python package.

[4] U.S. Securities and Exchange Commission. EDGAR application programming interfaces. https://www.sec.gov/search-filings/edgar-application-programming-interfaces, 2026.

[5] Alpaca Markets. Market data API. https://docs.alpaca.markets/docs/about-market-data-api, 2026.

[6] Center for Research in Security Prices. CRSP US stock database. Wharton Research Data Services, https://wrds-www.wharton.upenn.edu, 2026. Daily stock file and S&P 500 constituents to 2025-12-31.

[7] Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020. doi: 10.1093/rfs/hhaa009.